

Variation in Many-Analyst Projects: Harmful or Helpful?

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- ▶ Many-analyst studies **describe** but **do not quantify** decision sensitivity.

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- ▶ 16 teams, 6 decisions

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Quantify the relative contribution of harmful vs. helpful decisions to effect size variation.

Overall Multiverse Results

- ▶ 4,913 analyses
- ▶ Effect sizes ranged from **0.71** to **1.68**.

Effect Size	Percentage
Significantly negative (RR < 1)	65.3%
Significantly positive (RR > 1)	14.3%
Non-significant	20.4%

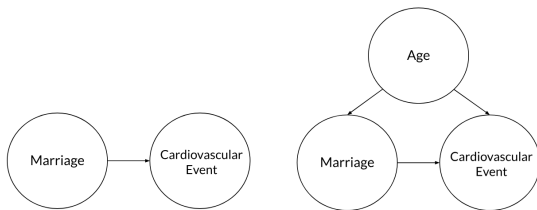
- ▶ Different analytical pathways can lead to **different conclusions**.

Decision Sensitivity

- ▶ **Statistical model choice** → harmful
- ▶ **Adjustment set** → helpful

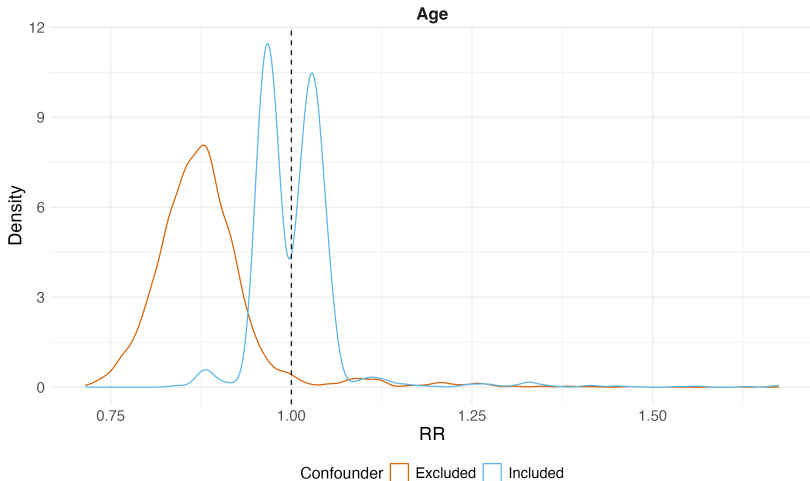
Decision Sensitivity

- ▶ **Statistical model choice** → harmful
- ▶ **Adjustment set** → helpful
- ▶ Sensitivity to **age** and **sex adjustment** reflect meaningful differences in what was estimated, not arbitrary noise.



Density plots of RR per adjustment variable

Each panel shows densities when including/excluding a specific adjustment variable



Note: Dashed line indicates $RR = 1$

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Thank you! Questions?

References

Kowall, Bernd, Linda Juel Ahrenfeldt, Jale Basten, Heiko Becher, Tilman Brand, Julia Braun, Swaantje Casjens, et al. 2025. "Marital Status and Risk of Cardiovascular Disease – a Multi-Analyst Study in Epidemiology." *European Journal of Epidemiology* 40 (5): 497–509.
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